Clustering Analysis of Employment in European Countries (1979)

1.Executive Summary

1.1 Goal and Background

The objective of this coursework is to identify patterns and similarities among the employment across the European countries in 1979 by examine the percentage of employment in various sectors such as agriculture, manufacture, finance and other services, we aim to uncover the underlying structures in the economies of these countries.

1.2 Approach

The data set is consist of the employment percentage in different sectors across different countries, to better comparability we have to standardize the data first.

Clustering methods were used to group the countries:

- K-means clustering
- Hierarchical clustering

The number of clusters was determined by elbow method and silhouette method, found K= 3 as an appropriate number of the cluster.

1.3 Major Findings

Three clusters were identified.

- **Cluster 1 and Cluster 2** contain most of countries with balanced employment profile which is closer to the average and slightly higher in the manufacture sector.
- **Cluster 3** contains only 2-3 countries with very high agricultural employment (41%–67%) and very low shares in manufacturing, construction, services, and transport.

The strong agreement between k-means and hierarchical methods confirms the robustness of these patterns. Table below shows the k-means cluster summary:

Cluster	Agr	Min	Man	PS	Con	SI	Fin	SP	TC	Interpretation
1	+0.35	+0.95	+0.39	+0.15	+0.11	-0.95	-1.03	-0.41	+0.51	**Mining + some industry**
2	-0.49	-0.45	+0.07	+0.03	+0.27	+0.67	+0.42	+0.41	+0.01	**Service-oriented / Modern economies**
3	+2.48	-0.16	-2.09	-0.82	-2.62	-1.56	+0.78	-1.67	-2.12	**Agriculture-dominant**

This illustrates a huge divide among European countries in 1979, a small group with large agriculture sector, transition group with balance of everything and other with more modern and service-oriented industries.

2. Main Report

2.1 Data Description

The dataset is the percentage of the various sector employment in different European countries in 1979. It has 10 variables (countries, agriculture, mining, manufacturing, service industries, construction, power supply, SP and TC), Each row represents a countries and percentage of employment in different sectors.

2.2 Methodology

Before clustering, the data has been scaled to eliminate the effect of different scale. A distance matrix had been made and plot it as heatmap to know if there are any similarities among the employment of the different countries shown in the figure1below:

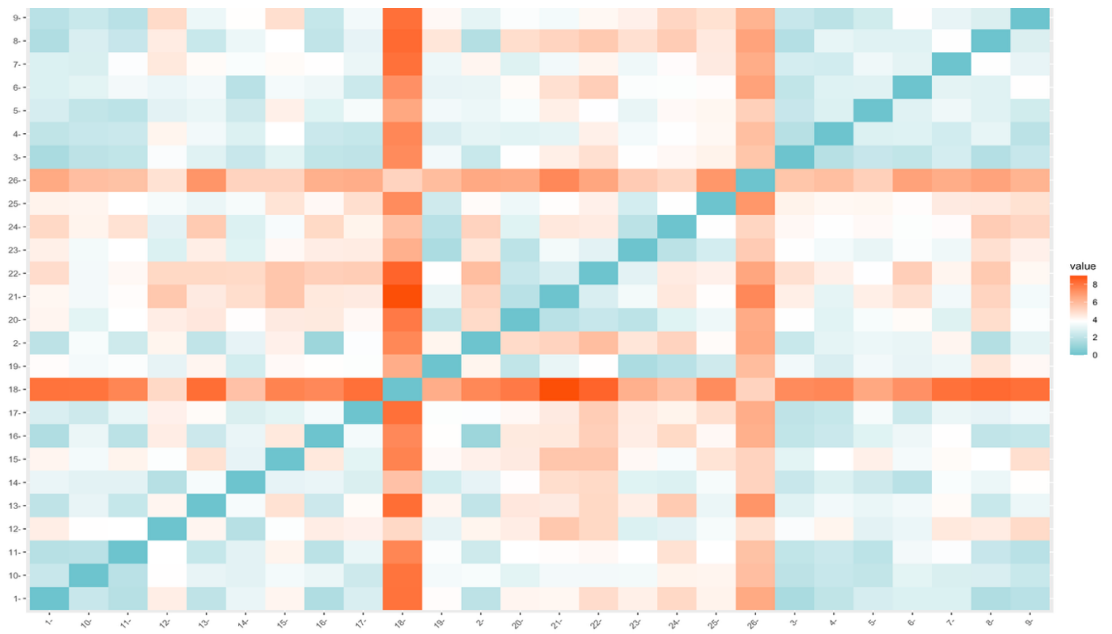

Figure 1

Heatmap of the distance matrix illustrating most of the observations are very close to each other and reveals noticeable patterns of similarity among countries. Darker colours indicate greater dissimilarity, while lighter colours indicate higher similarity. For example, those observations in the top right rectangle may be quite similar to each other and may be falls in similar cluster.

Two clustering methods were used:

- K-means clustering
- Hierarchical clustering using Ward.D2 method

The number of clusters was determined using the elbow and silhouette method, which showed a clear reduction in within-cluster variation up to k = 3, after which improvements became marginal shown in the figure 2 and 3 below:

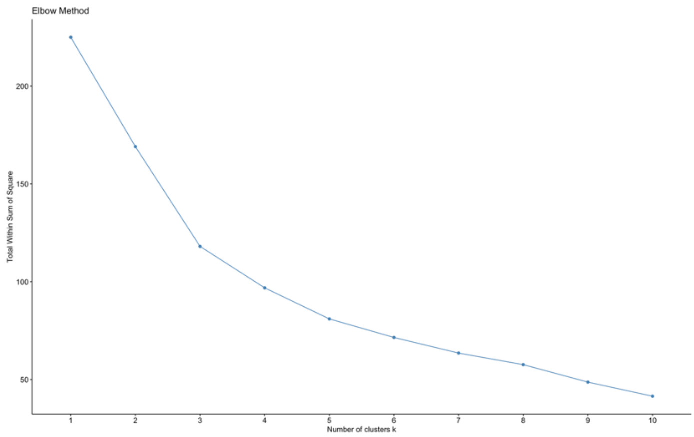

Figure 2

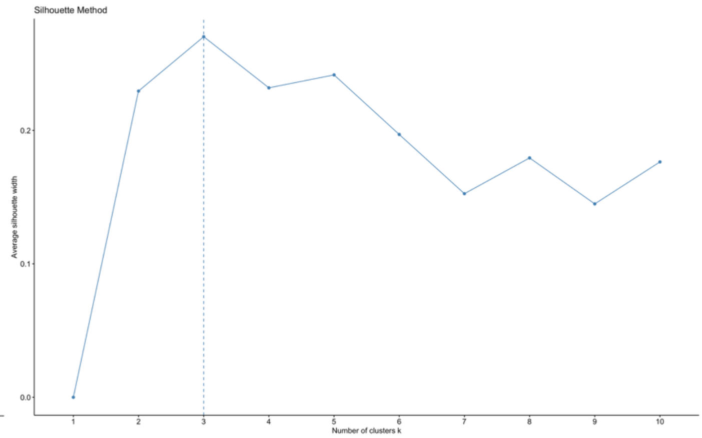

Figure 3

2.3 Results

2.3.1 K- means clustering

The K-means clustering results gives three groups of countries, each representing a different economic structure shown in the figure 4 below:

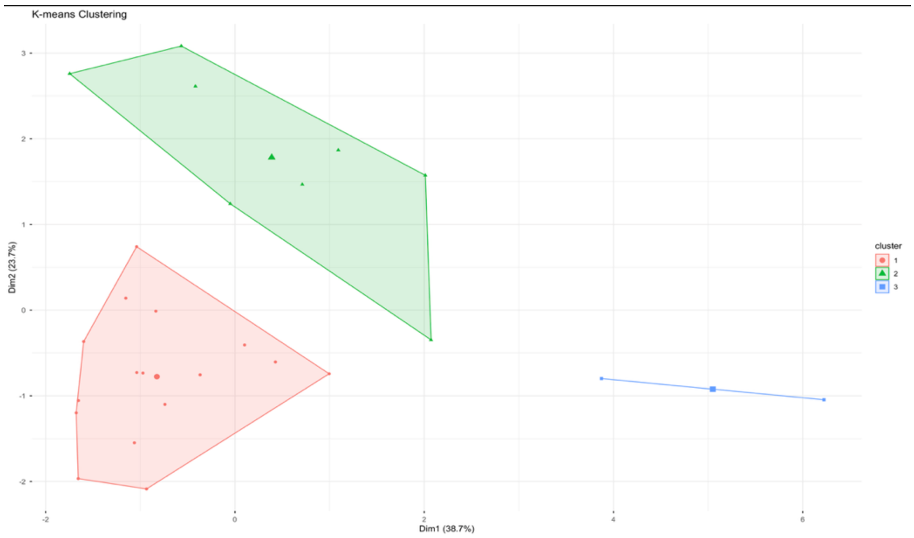

Figure 4

Table below contain countries in each cluster:

Cluster	**Countries**
Cluster 1	Greece, Bulgaria, Czechoslovakia, East Germany, Hungary, Poland, Rumania, USSR
Cluster 2	Belgium, Denmark, France, West Germany, Ireland, Italy, Luxembourg, Netherlands, UK, Austria, Finland, Norway, Portugal, Spain, Sweden, Switzerland
Cluster 3	Turkey, Yugoslavia

Cluster 1: Transitional Economies

This cluster is characterized by moderate employment in agriculture and manufacture (+0.35 and +0.39 respectively from average) and high employment in mining (+0.95 from avg.), with relatively very lower levels in services and finance (-0.95and -1.03 form avg.). These countries appear to be transitioning from agriculture-based economies toward more industrialized based economies, not yet developed strong service sectors.

Cluster 2: Industrial and Service-Oriented Economies

Countries in this cluster show low agricultural employment (-0.49 from avg.) and better combination of moderate employment in construction (+0.27 from avg.) and finance (+0.42 from avg.) and power supply (+0.41 from avg.). Additionally, this group has the high levels in services industries (+0.67 from avg.). These characteristics indicate advanced, diversified economies with strong industrial and service sectors.

Cluster 3: Agricultural Economies

This cluster is dominated by agriculture, with a very high average employment share (+2.48 from avg.). Employment in finance (+0.78) which is significantly higher but very low in other sectors like manufacture (-2.09 from avg.), construction (-2.62 from avg.) and service (-1.56 from avg.). These countries are less industrialized and rely heavily on agriculture sector.

2.3.2 Hierarchical clustering

Hierarchical clustering was conducted on the standardized employment data using WardD2 linkage method. The resulting dendrogram (Figure 5) shows a clear structure with a large height jump which indicates strong separation between a small group of countries and the rest of countries.

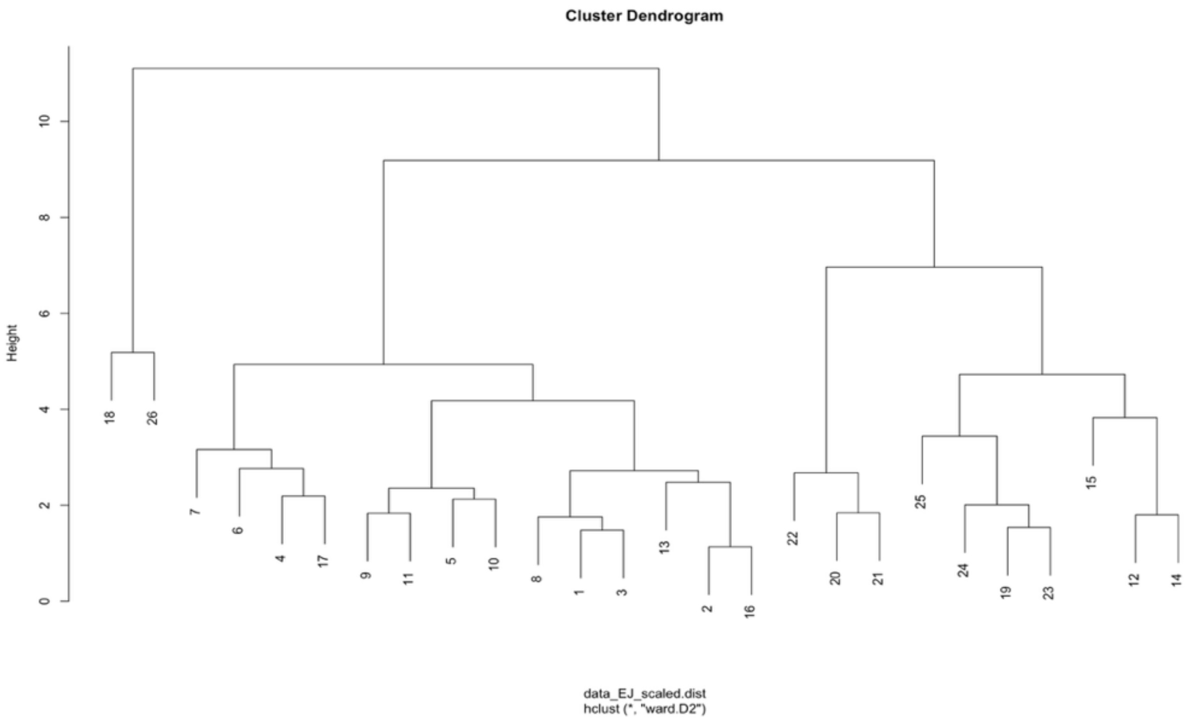

Figure 5

The table below contain the countries in each cluster :

Cluster	Countries
Cluster 1	Belgium, Denmark, France, West Germany, Ireland, Italy, Luxembourg, Netherlands, UK, Austria, Finland, Norway, Sweden, Switzerland
Cluster 2	Greece, Portugal, Spain, Bulgaria, Czechoslovakia, East Germany, Hungary, Poland, Rumania, USSR
Cluster 3	Turkey, Yugoslavia

The cluster plot (Figure 6) visualizes the separation in the first two principal components. Cluster 3 appears as a distinct outlier on the left which only 2 countries, while Clusters 1 and 2 occupy the central and right portions of the plot which got 14 countries in cluster 1 and 10 countries in cluster 2.

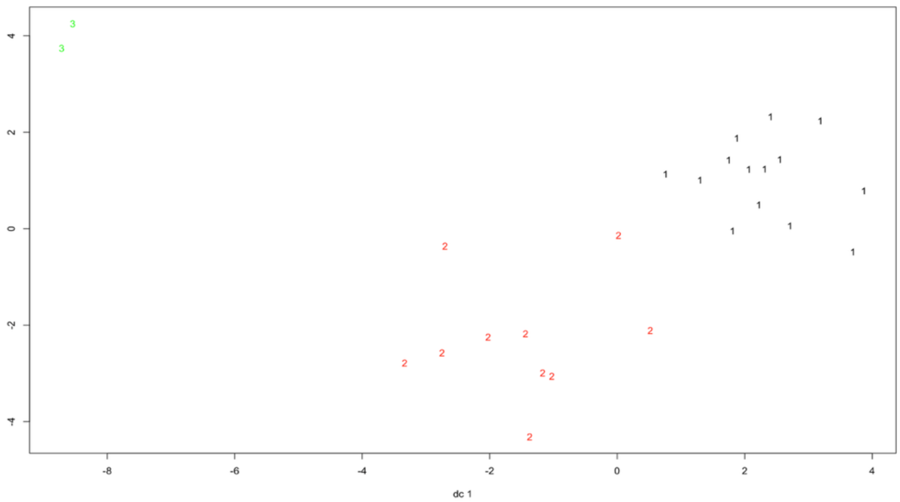

Figure 6

Cutting the dendrogram at k=3 produced the same three clusters as k-means, with Cluster 3 again appearing as the clear agriculture-dominant outlier.

3 Conclusion

The clustering results reveal a clear economic structure across European countries. The three clusters represent different stages of economic development:

- Agriculture-dominated economies
- Transitional economies
- Modern and service-oriented economies

This shows the progression of economies, where countries evolve from primary sector dominance to industrialization and service-based economies.